



Problem Card Duel

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Difficulty: medium

Time Limit: 3 s

Description

Ali and Akram are playing a fun game with number cards. Ali gives Akram two groups of number cards — let's call them Group A and Group B (each group is just an array of numbers).

Akram's challenge is to make the numbers in Group A exactly match those in Group B, position by position. But there's a twist: he can only use one special move, called the "twist" operation.

There are two types of moves he can use on any card:

- Replace a card in Group A with $\text{twist}(A_i)$
- Replace a card in Group B with $\text{twist}(B_i)$

Where $\text{twist}(x)$ means: the largest proper divisor of x (i.e., the greatest number less than x that divides x exactly).

For example:

- $\text{twist}(12) = 6$



- $\text{twist}(18) = 9$
- $\text{twist}(1) = 1$ (because 1 has no proper divisors)

Akram can apply as many of these moves as needed on any of the cards in either group. His goal is to make both groups identical (i.e., the numbers at each index in both groups must be

the same).

Ali will only reward Akram with a brand-new game console if he completes this challenge in the minimum number of moves possible.

For each round of the game (test case), determine the minimum number of operations

Akram needs to make Group A and Group B equal.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

Each test case has 3 lines:

- The first line contains a single integer n ($1 \leq n \leq 10^5$) — the length of the two arrays.
- The second line contains n integers $A_1, A_2, \dots, A_n$ ($1 \leq A_i \leq 10^5$) — elements of the array A .
- The third line contains n integers $B_1, B_2, \dots, B_n$ ($1 \leq B_i \leq 10^5$) — elements of the array B .



Output

- For each test case, print one integer — the minimum number of operations (possibly zero)
- to make arrays A and B equal.

Examples

Input	Output
3	7
4	5
4 5 12 1	2
8 3 9 2	
2	
5 4	
10 6	
1	
2	
8	